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MILITARY SPECIFICATION

WARHEAD, 5-INCH ROCKET, SMOKE, WP, MK 39, MOD 0; FILLING ASSEMBLY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the white phosphorous (WP) filling and assembling of the Mk 39 Mod 0 rocket warhead, without explosive components.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
JAN–P–215 —Phosphorus, White.
MIL–G–16908 —Grease, Bearing (for General Ordnance Use).

STANDARDS

MILITARY
MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

DRAWINGS

DEPARTMENT OF THE NAVY—BUREAU OF NAVAL WEAPONS
289024 —Warhead, 5-Inch Rocket, Smoke, Mk 39 Mod 0—List of Drawings, Subassemblies, and Parts (2 sheets).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION
49 CFR 71–90—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (Revised 1956) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–90 (Rev. 1956).")

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All materials cited on Drawing 289024 or on the subsidiary drawings shall conform to the specifications thereon or the specific characteristics set forth on the drawings.

3.1.2 *Components.* All components of the warhead shall conform to the specifications and drawings listed on Drawing 289024 and subsidiary drawings.

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3.2 Rust prevention of components. Empty warheads and burster wells shall be stored with the rust prevention protection they had when they were received.

3.3 Filling. The warhead, with shipping plug removed and free from foreign matter, shall be filled with molten white phosphorus (WP) conforming to Specification JAN–P–215 (see 6.2). When process water is used in WP filling lines, the water shall be buffered by a suitable alkaline salt to maintain pH value between 6.5 and 8.0 (see 3.4).

3.4 Filling height. The surface of the WP in the filled and closed warhead shall be 3.50 to 4.03 inches below the nose of the burster well. A maximum height of ¼ inch of water shall be permitted above the WP. These requirements are applicable when the warhead is inspected in a vertical noseup position and when tested as specified in 4.5.1 (see 6.3).

3.5 Assembly. Closing, coating, marking, and assembling of the warhead shall be as specified on Drawing 1380986, which is listed on Drawing 289024. All threads in the assembly shall be coated with grease conforming to Specification MIL–G–16908 after the warhead has been tested for leakage. The fuze adapter (Dwg. 433698), the gasket (Dwg. 483488), and the shipping plug (Dwg. 434039) shall be assembled to a seated hand-tight fit.

3.6 Cleaning after closing. The filled and closed warhead and the interior of the burster well shall be cleaned to remove any grease, oil, WP, or other foreign matter (see 6.4).

3.7 Leakage. The cleaned warhead shall not leak when tested as specified in 4.5.2.

3.8 Solidification. The filled and closed warhead shall be placed in a noseup position until the WP has solidified.

3.9 Workmanship. The filled, closed, and assembled warhead shall be free from damage which would adversely affect the assembly of the next components or damage which would affect functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all the requirements in section 3 and section 5 have been met (see 6.6).

4.3 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 *Lotting.* A lot shall consist of the warheads filled and assembled by one manufacturer, at one filling installation in not more than one work shift, from the same materials, under essentially the same filling and assembling conditions, and offered for delivery at one time. However, not more than one lot of empty warheads or burster wells used in a continuous run shall be represented in any lot of filled and assembled warheads.

4.4.2 *Sampling for nondestructive examination.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.3 *Examination.* Sample warheads and the level A preparation for delivery requirements shall be examined in accordance with the classification of defects and Standard MIL–STD–105.

4.4.4 *Classification of defects.*

4.4.4.1 *Warhead, smoke, 5-inch rocket, Mk39 Mod 0, WP filling assembly and marking diagram (Dwg. 1380986).*

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Categories	Defects	Acceptance standards
Critical[1]		
	1. Leakage.	4.5.2
Major—AQL 1.0 percent defective:		
	101. Component missing or incorrectly assembled.	
	102. Filling excessive (WP or water).	4.5.1
	103. Burster well adapter thread damaged.	
	104. Base thread damaged.	
	105. Marking incorrect.	
	106. Burster well adapter not seated.	
	107. Weight incorrect.	
	108. WP contamination (see 3.6).	
Minor—AQL 2.5 percent defective:		
	201. Coating missing, inadequate, or incorrect.	
	202. Marking missing or illegible.	
	203. Filling insufficient.	4.5.1
	204. Foreign matter, except WP (see 3.6).	

[1] Each warhead in each lot shall be examined.

4.4.4.2 *Preparation for delivery (section 5).*

Categories	Defects
Critical—None defined.	
Major—AQL 4.0 percent defective:	
	101. Pallet incorrect.
	102. Quantity per pallet incorrect.
	103. Strapping incorrect.
	104. Damage (pallet wracked or broken).
	105. Marking incorrect, missing, or illegible.

4.5 Tests.

4.5.1 *Filling height.* The warhead shall be radiographed (see 6.1) while the temperature of the WP filling is 130°±10°F. to satisfy the requirements of 3.4.

4.5.2 *Leakage.* Following the final cleaning operation and while the filling is still molten, the filled and closed warhead shall be placed in a nosedown position or rolled on its side for a minimum period of 1 minute and observed for emission of smoke.

5. PREPARATION FOR DELIVERY

5.1 Level A packing. The filled and closed warheads shall be packed for shipment or storage on the same pallets as the empty warheads were received. The pack shall comply with Interstate Commerce Commission regulations.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129 and Interstate Commerce Commission regulations.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The supplier will not be required to furnish radiographic equipment.

6.2 Precautions. Because of the hazardous nature of filling munitions with WP, filling installations are advised to take all necessary precautions to protect personnel from injury while filling in accordance with this specification.

6.3 Filling aid. Warheads filled with the surface of the WP 3.94 to 4.31 inches below the nose of the warhead and the surface of the water at a maximum height of ¼ inch above the WP have been found to comply with 3.4 provided the internal configurations of the warheads are within the tolerances specified on the drawings. The WP in the warheads will weigh approximately 10½ pounds. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Decontamination. Warheads may be decontaminated by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance and a power driven wire brush.

6.5 Safety classification. This item is group C chemical ammunition and carries the following Ordnance Corps safety classification: storage compatibility group A, explosive hazard class 2.

6.6 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications,

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Army—Chemical Corps

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